

UNIT 2

The ecosystem is highlighted here because it is by learning how it functions that we can also learn and adapt for the survival of mankind on earth



Content

- Laws of matter and Energy Laws
- Ecology
- Ecosystems
- Biogeochemical cycles
 - Hydrological cycle
 - Gaseous cycles
 - Sedentary cycles

Energy principles and laws of Energy

- Matter is anything that has mass and occupies space
- Energy is what is used by all living things to move matter around and to change matter from one form to another
- Work involves changing energy from one form to the other
- There are 3 major laws operating in ecosystems

The Laws of matter and energy

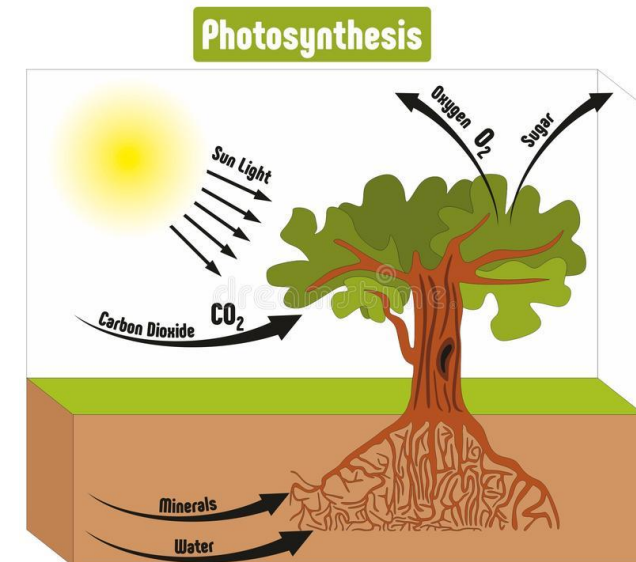
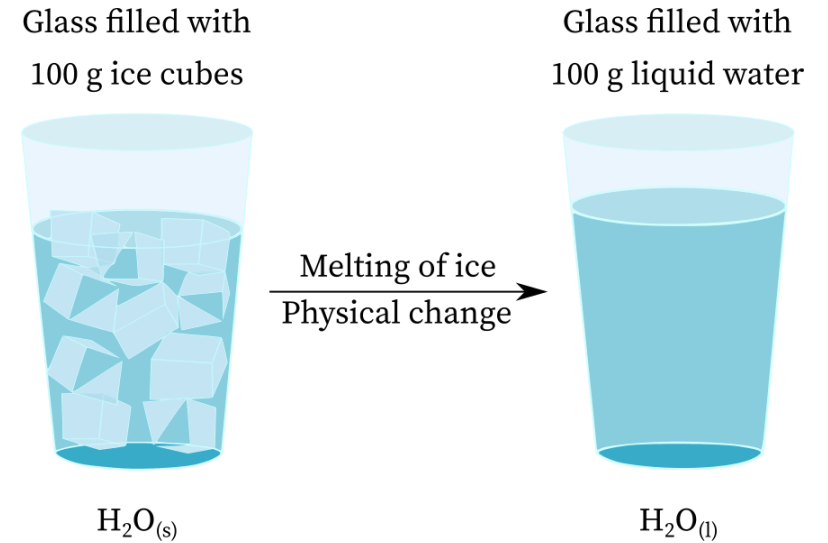
- The Law of Conservation of Matter
- The 1st Law of Energy or Thermodynamics
- The 2nd Law of Energy or Thermodynamics

The Laws of matter and energy

- The Law of Conservation of Matter- In any ordinary physical or chemical change, matter is **neither created nor destroyed but merely changed** from one form into the other
- The 1st Law of Energy or Thermodynamics –In any ordinary physical or chemical change, energy is **neither created nor destroyed but merely changed** from one form into the other.
- The 2nd Law of Energy or Thermodynamics-The quality of energy you put in a systems is always **degraded** into a low quality of energy usually in the form of heat. The entropy of the universe is always increasing

The Law of the conservation of matter

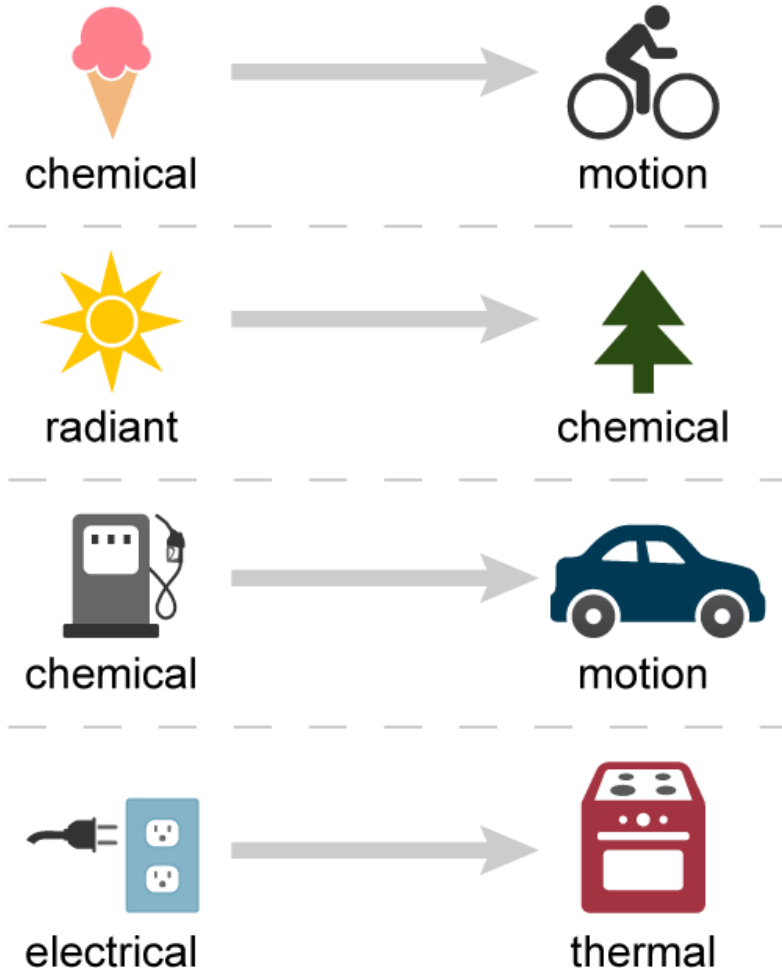
- The law states “In any ordinary physical or chemical or physical or chemical change, matter is neither created nor destroyed but merely changed from one form into another”.
- This means that **whatever we think we have thrown away exists with us in one form or the other**



The Laws of Energy

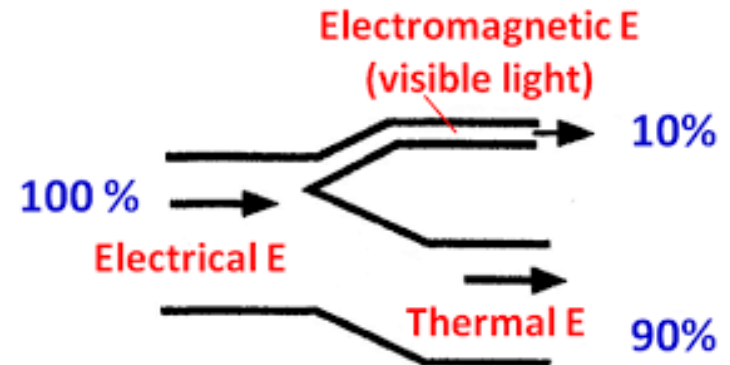
- First law states that in any physical or chemical change, energy is neither created nor destroyed but merely changed form one form to the other
- Means we need energy to get energy

Energy transformations



The Laws of Energy

- The second law states that the quality of energy you put in a system is always **degraded** into a low quality of energy which is mostly in the form of heat
- This energy cannot be used to do useful work and is called **entropy**
- The entropy of the universe is thus increasing



1st and 2nd Laws of Energy

- Normally the quality of the energy available for useful work will always be lower in quality than the initial energy.
- It is unfortunate that high use of resources particularly petroleum products leads to increase in atmospheric temperature because the petroleum is reduced into low quality energy in the form of heat anytime it used.
- We can recycle matter but for all practical purposes now **we cannot recycle energy.**

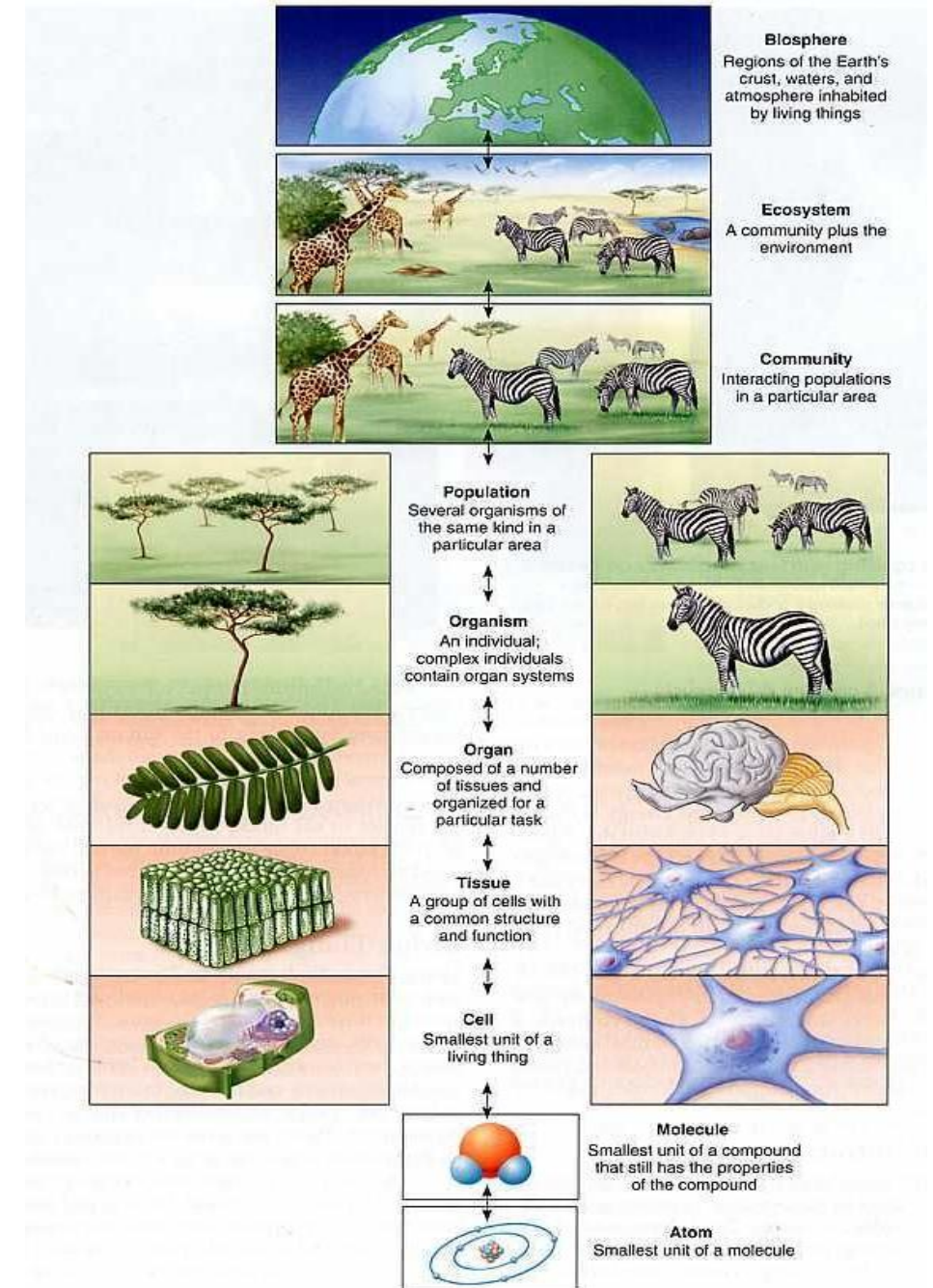
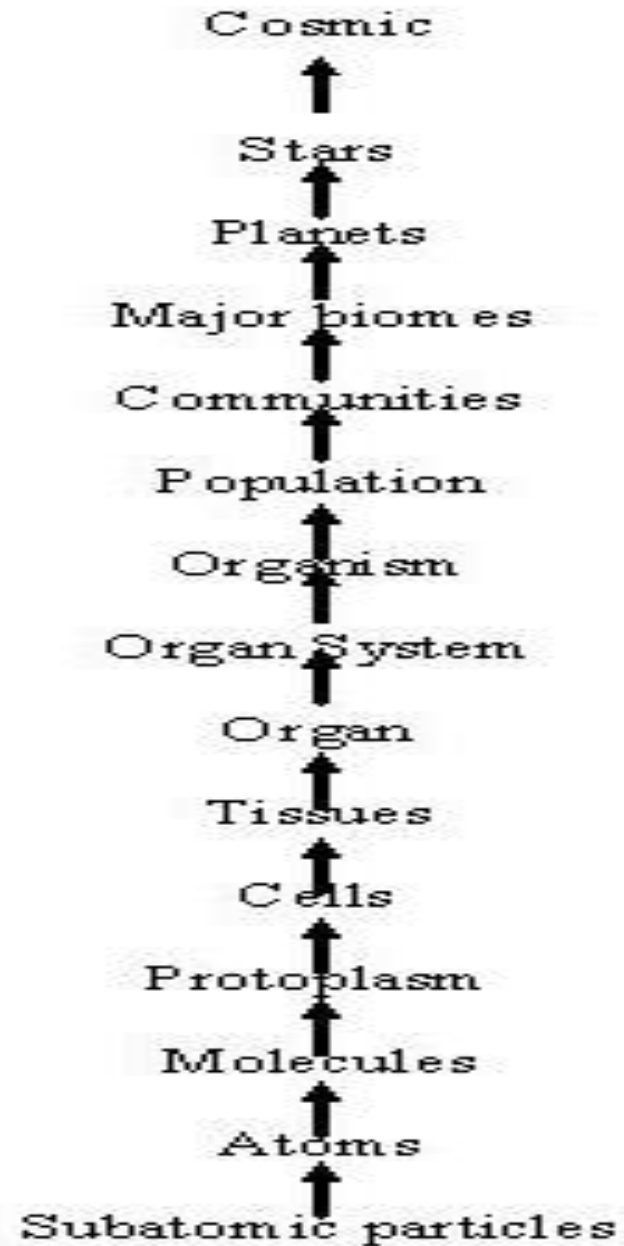
1st and 2nd Laws of Energy

- There is an increase in entropy.
- Eventually, non living objects wear out and living things die and decompose.
- This process of becoming more disordered coincides with the constant flow of energy towards a dilute form of heat.
- This dissipated, low quality heat has little value to us, since we are unable to use it.

1st and 2nd Laws of Energy

- It is important to understand that energy that is of low quality from our point of view may still have significance to the world in which we live.
- For example the distribution of heat energy in the ocean tends to moderate the temperature of the coastal climates.

Levels of Organisation of Matter

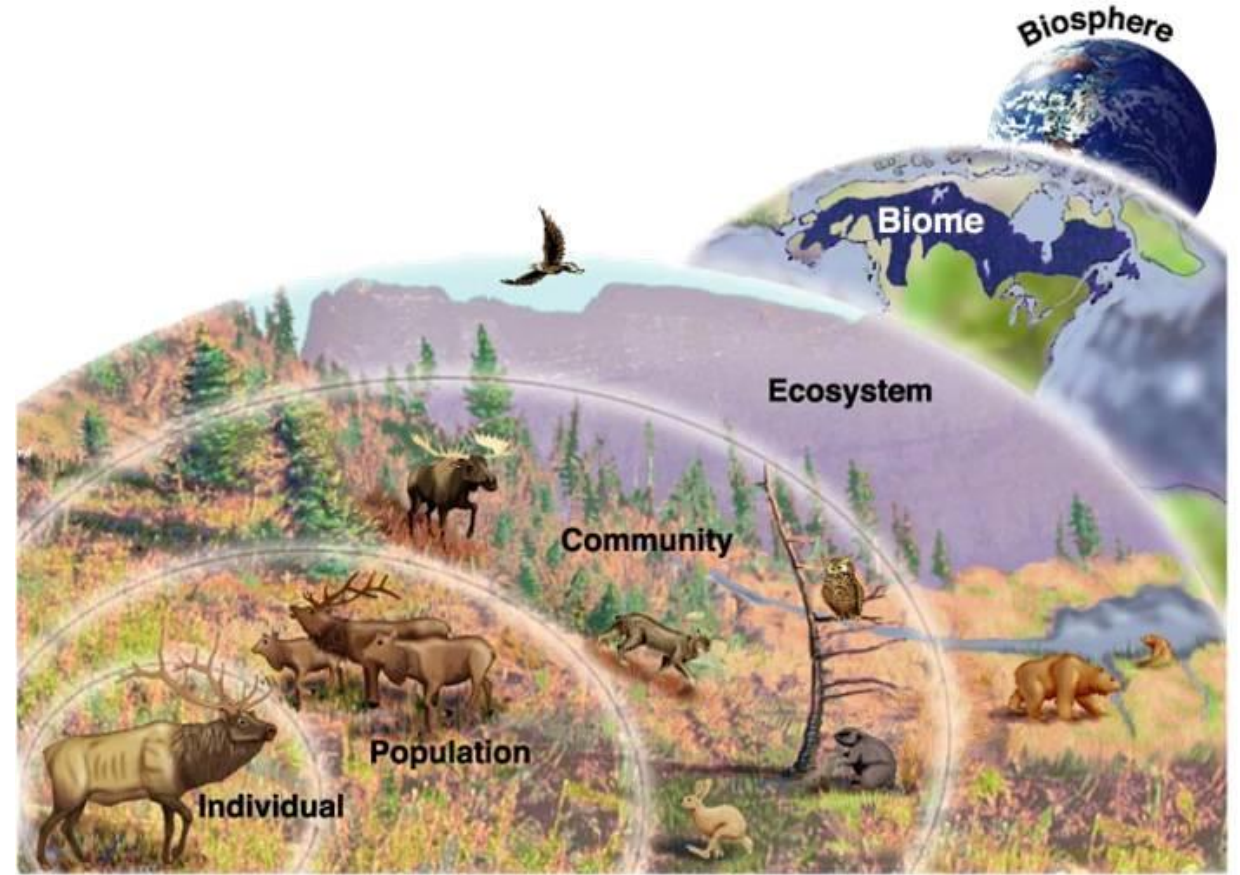


Ecology

Ecology – study of the relationship between living organisms and their environment.

Ecology- The study of the structure and function of nature.

There are **6** levels of organisation in ecology.



Organism < **Population** < **Community** < **Ecosystem** < **Biome** < **Biosphere**

Basic Ecological Concepts

- ◎ **Environment**- An aggregate of all the external conditions that tends to affect organisms both physical and biological.
- ◎ **Ecosphere/Biosphere** - The sum total of all the ecosystems and their interactions in the world. The part of the earth where life exists.
- ◎ **Biome** – A major community of plants and animals occupying a large geographical area adapted to a particular climatic or environmental conditions such as precipitation and temperature. Eg are forest, grasslands, tundra, marine, freshwater and deserts.

Ecosystem

- **Ecosystem** – A self-sustaining and self-regulation of a community of living organisms living together and interacting with their environment.
- A community of organisms and their abiotic environment
- Involves the movement of energy and materials through the communities
- Ecosystems have no definite boundaries
- Egs. Forests, a lake with its biota (Flora and fauna), fallen log, pond.
- Have a **continuous input of energy** to retain its stability.
- The only significant source of energy for most ecosystems is **sunlight**.

Lake



Forest

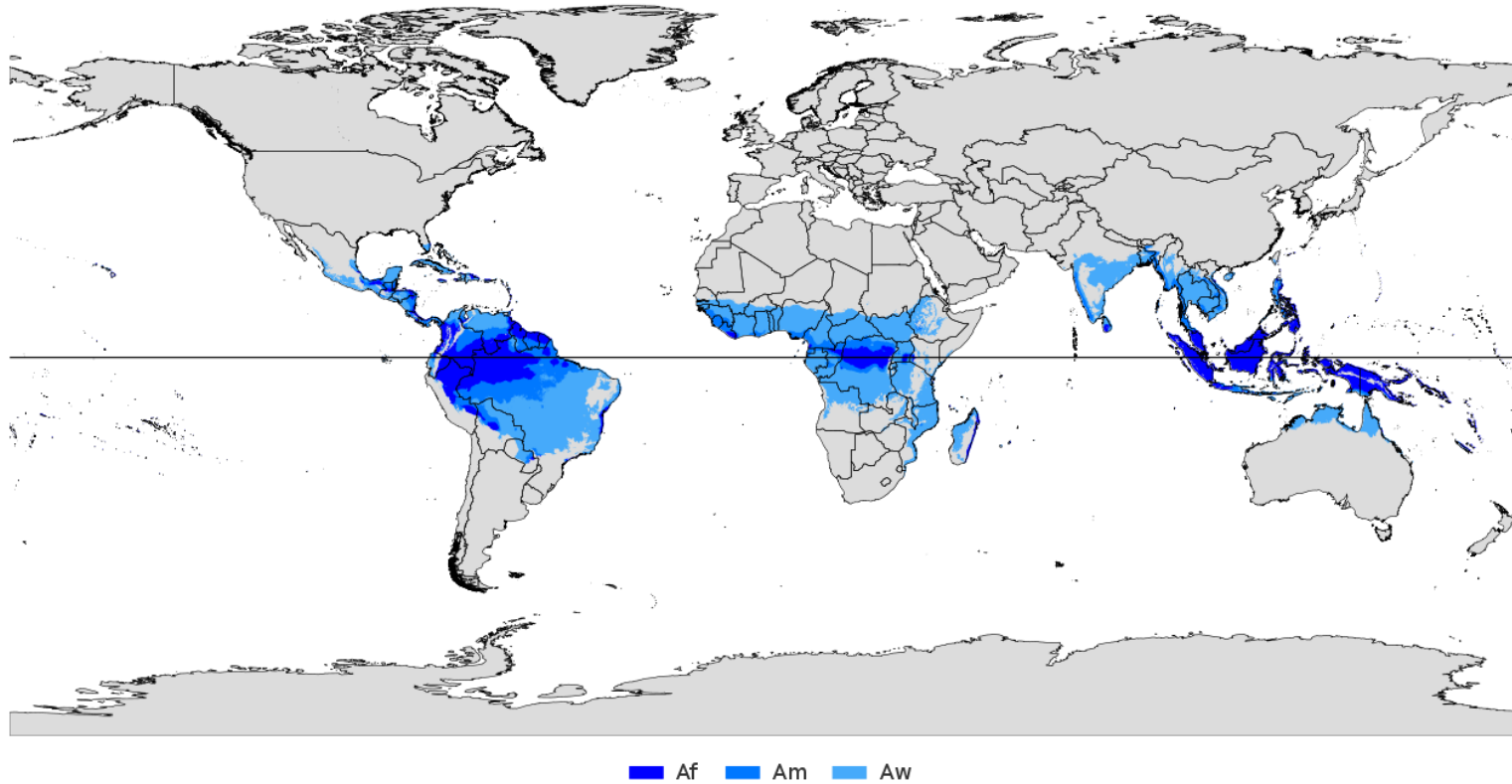


Desert



Tropical ecosystem

Tropical (1980-2016)



Tropical ecosystems

- Tropical climates are found around the equator - Latitude 25 degrees south and 24degrees North
- Temperatures of 18 degrees Celsius or higher
- Characterised by small temperature changes and long summers.
- Due to the high temperature and abundant rainfall, some plants can grow throughout the year.
- High temperature and humidity is the most suitable environment for epiphytes to grow. Plants of all sizes can vegetate under tropical climates.
- Vegetation grow in layers: [shrubs](#) under tall trees, and bushes under shrubs. Almost every inch of space is being well used.
- Tropical plants are rich in resources, including coffee, cocoa and oil palm.
- Has a very high species diversity index and support several plant and animal species hence the prevalent of several diseases.

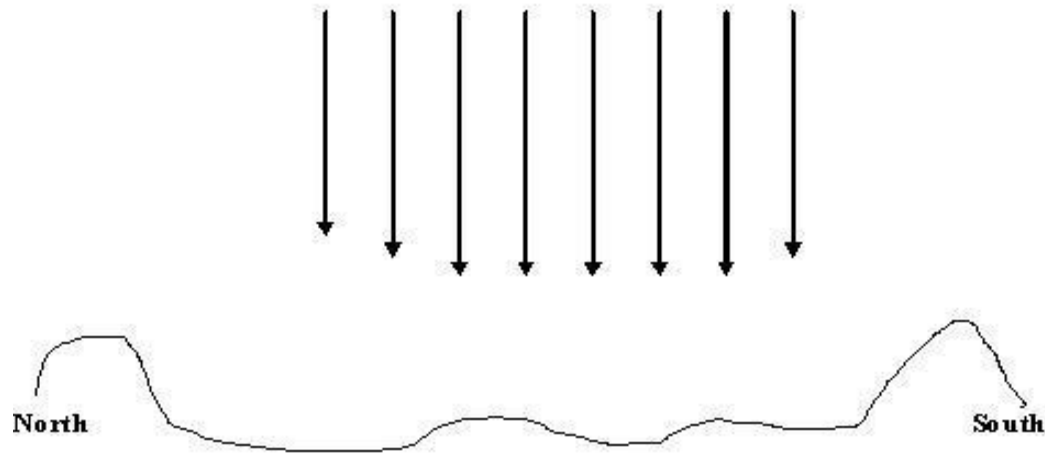
The Tropical Environments

- Tropical forest of standing trees
- Savannah grasslands and woodlots
- Deserts

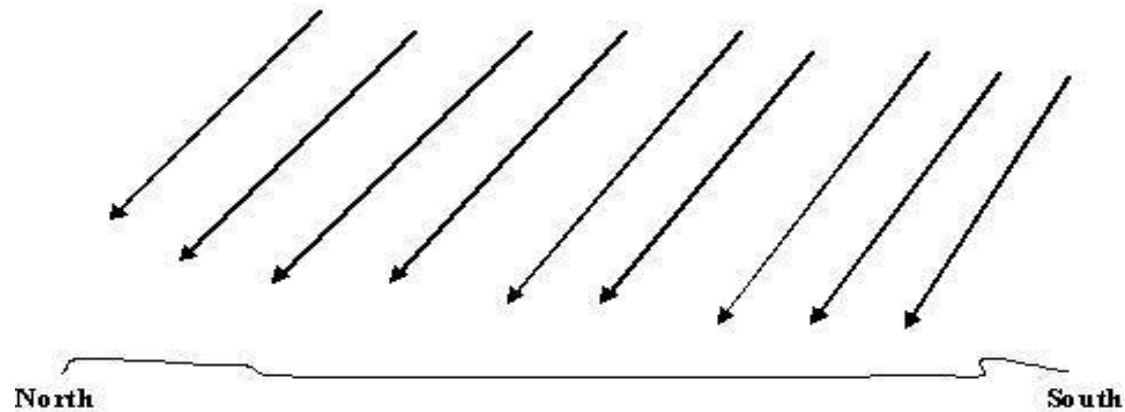


Uses of the Tropical Ecosystems

- Tropical crops
- Sequesters of carbon dioxide
- Medicinal use
- Soil conservation
- Timber
- Fuel
- Tourism
- Transpiration
- Clothes and Food
- Biological diversity
- Odour absorbing plants



The sun's rays in the tropics are at right angles whereas it is slanted in the temperate zones.



Light rays from Sun in Temperate Latitude of Northern and Southern Hemisphere

Ecosystem

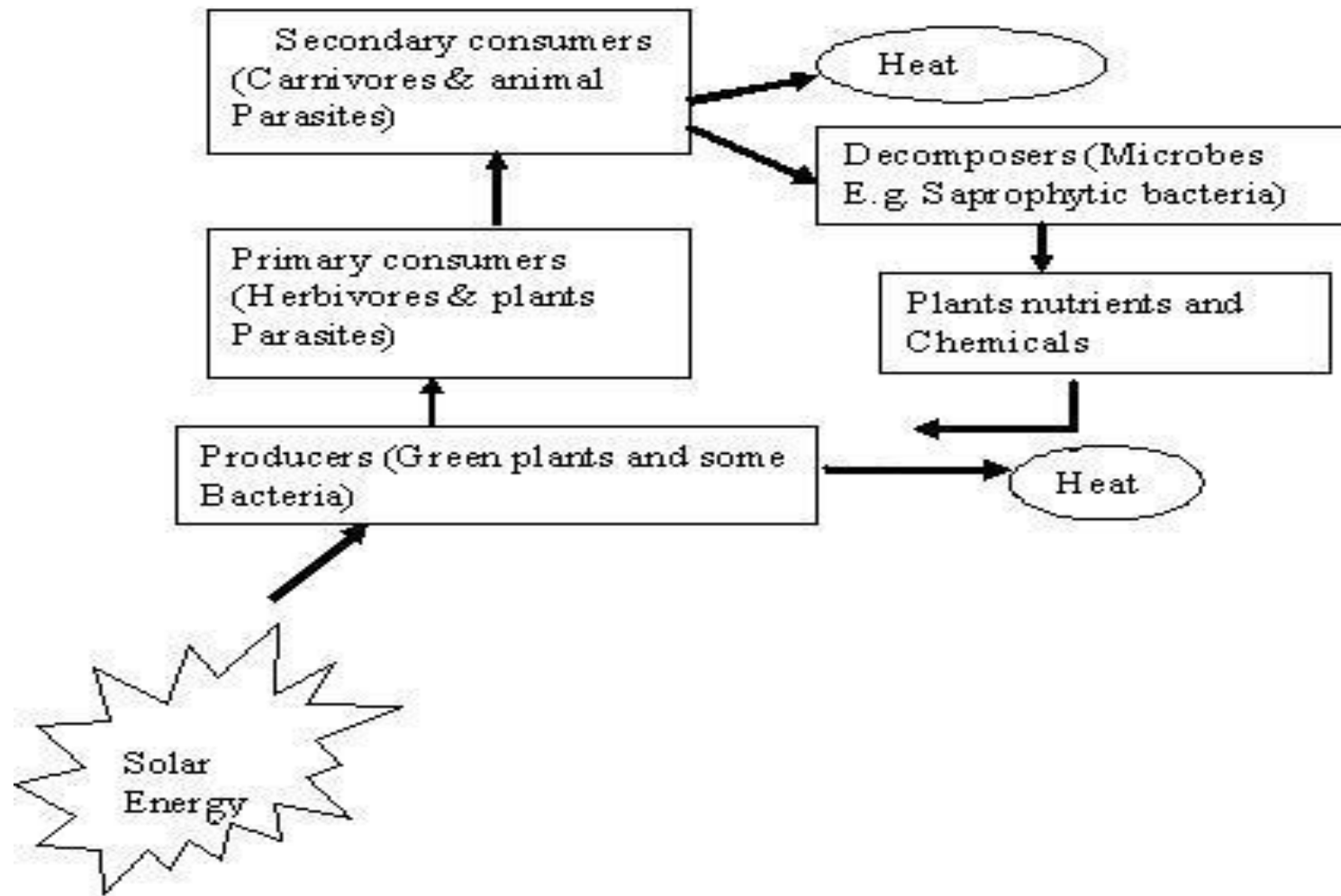


Figure 3: Basic Components of the Ecosystem

Community, Population and Organism

- **Community** is a group of plant and animal populations living and interacting in a given locality.
- **Population** is a group of plants or animals of the same species in a given locality at a given time.
- **Organism** is an individual living thing



Ecosystem

Community

Population

Organism

Ecological Niche and Habitat

- Ecological niche is a species total **role and function** in its environment and the **resources** it requires.
- It encompasses the environmental conditions necessary for its survival and its ecological role in an ecosystem.
- How a species interact with both the biotic and abiotic environment.
- Its effects on other organisms with which it interacts
- It may include where it lives or what it feeds on
- The **habitat** of an organism is the space that the organism occupies/ the place where it lives.
- It may include influences of biotic and abiotic conditions
- Where it finds its food, mate etc.



Ecosystem Structure

The ecosystem has two major components:

- i. The non-living (abiotic) part
 - ii. The living or biotic part
- The structure of the ecosystem is directly related to energy transfer and matter recycling.

Living (Biotic) Portion of an Ecosystem

i. Producers (Plants or autotrophs):

- Plants, ranging from tiny floating phytoplankton (algae, diatom etc.) in water ecosystems to giant trees

ii. Macroconsumers (animals or heterotrophs):

- Organisms who cannot manufacture their own food and must consume the preformed food compounds found in plants and animals

iii. Decomposers (micro consumers or saprotrophs):

- Tiny organisms such as bacteria that break down the bodies and complex compounds in dead animals and plants into smaller substances

Roles of Organisms in Ecosystem

Producers (Autotrophs): produce their own food

- 2 kinds of producers- Phototrophs and chemo-autotrophs
- **Phototrophs:** They are mainly green plants. Also green algae and cyanobacteria
- They use simple inorganic substances in their environment
 - sunlight
 - Water
 - Carbon dioxide

in the presence of chlorophyll to produce / make complex organic molecules.

The process is photosynthesis.

Roles of Organisms In Ecosystem

- Chemo-autotrophs

- They are microorganism which can produce organic matter from chemicals
- Through the Oxidation of certain chemicals in the absence of sunlight
- The process is called chemosynthesis
- Eg. Is *Nitrosomonas* -uses ammonia as an energy source in the presence of CO₂ and oxygen.

Roles of Organisms In Ecosystem

- Consumers (Heterotrophs):
 - They cannot produce their own food so they feed on organic matter (source of food) to provide themselves with energy and the organic molecules necessary to **build their own bodies**.
 - They break down the organic matter to simple inorganic molecules.
 - There are different types based upon the way they obtain their food and the kinds of food they eat.

Roles of Organisms In Ecosystem

- a) **Primary consumers**;
 - **Herbivores**: are animals that eat producers (plants and phytoplanktons) as a source of food. Eg. whales
- b) **Secondary consumers**;
 - **Carnivores**: are animals that eat other animals. Eg. Frogs, lizard, hawks
 - **Scavengers** are types of carnivores that feed on dead animals eg. Vultures, catfish
 - **Omnivores**: animals that feed on plants and other animals eg. pigs, rats
- c) **Tertiary consumers**: carnivores or omnivores that eat other carnivores eg. Man or bald eagle.

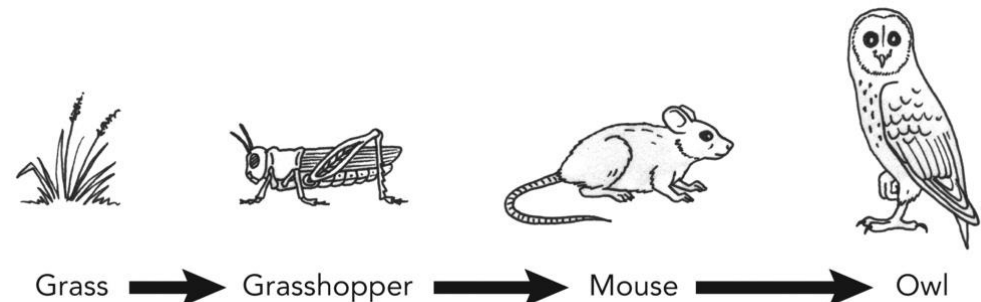
Roles of Organisms In Ecosystem

- d) **Detrivores**: feed on dead organisms and waste matter. Eg. Ants, millipedes, earthworms, termites, crab
- e) **Decomposers** (saprotrophs): organisms that break down (decompose) bodies and complex compounds in dead animals and plants into simpler substances for food. They use non-living organic matter as a source of energy and raw materials to build their bodies. Eg. Bacteria, protozoa and fungi.
- Whenever an organism sheds a part of itself, excretes waste products or dies, it provides a source of food for decomposers.
- Decomposers are extremely important in breaking down organic matter and releasing the organic matter for reuse by producers.

Food Chains And Food Webs

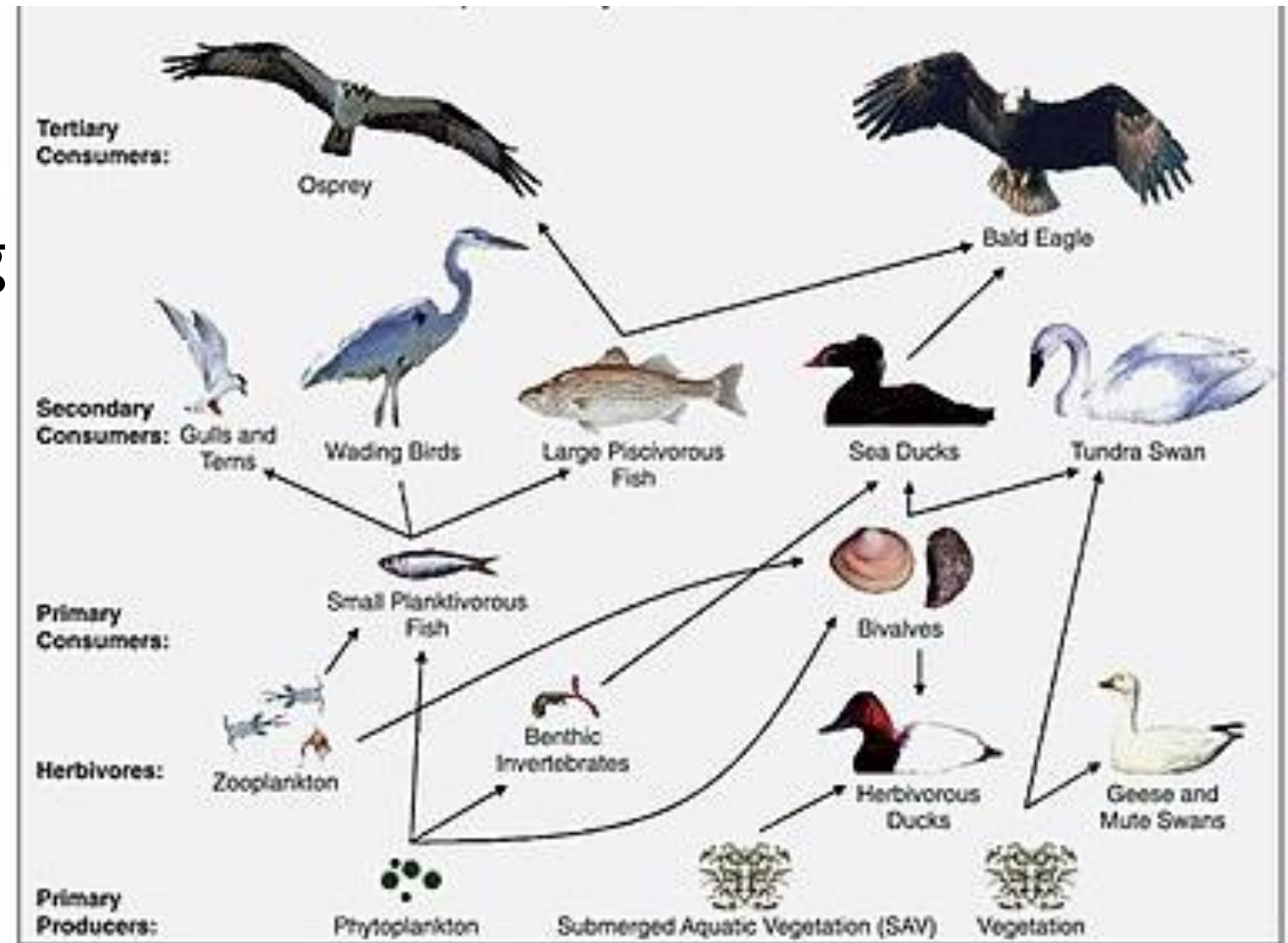
Food Chain

- A diagram showing one pathway of how energy flows in an ecosystem.
- A series of organisms occupying different trophic levels through which energy passes as a result of one organism consuming another.
- A food chain is simple and it shows that all life forms and all forms of food begin with sunlight and producers



Food Chains And Food Webs

- Food web
- It is a diagram showing the varying feeding relationships in an ecosystem.
- Shows the many pathways of energy flow in an ecosystem.

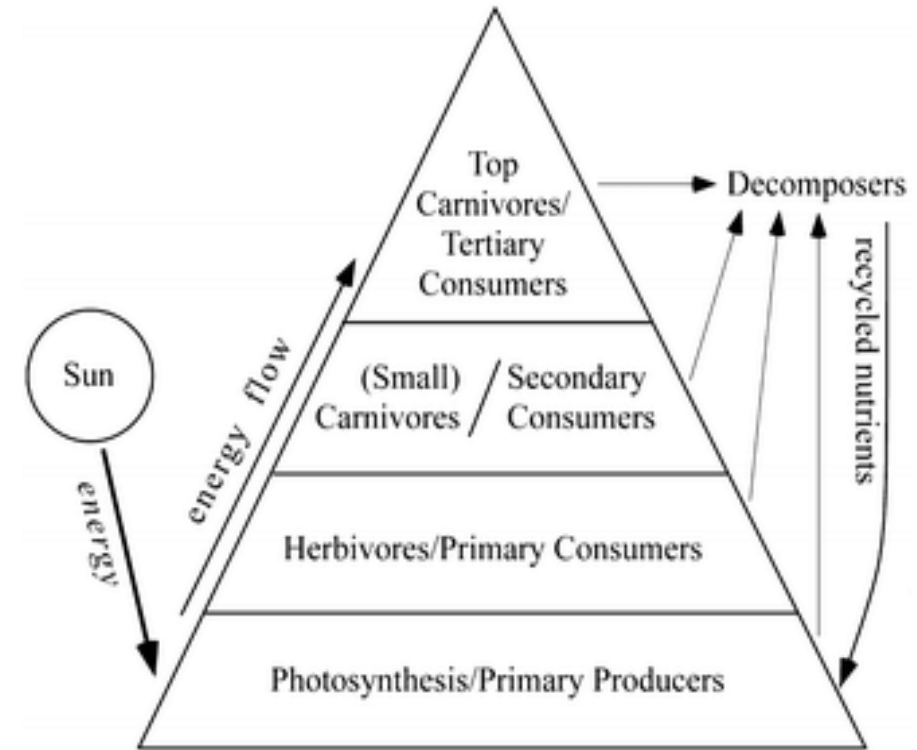


Trophic Levels and energy flow in Ecosystems

- Trophic structure – feeding levels in an ecosystem
- Each step in the flow of energy through an ecosystem is known as a trophic level.

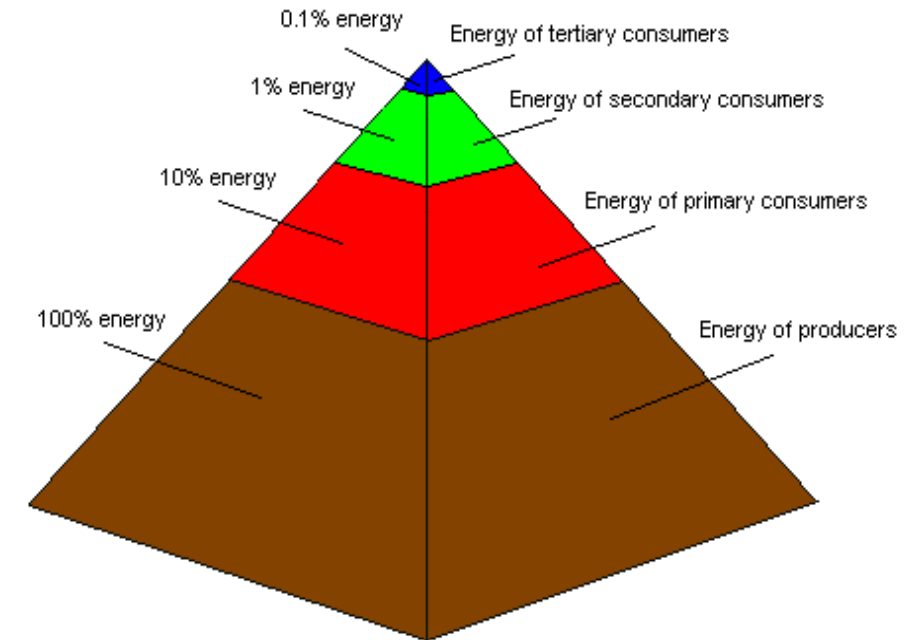
- Producers (Autotrophs) constitute the first trophic level
- Herbivores - second trophic level.
- Carnivores that eat herbivores - third trophic level
- Carnivores that eat other carnivores - fourth trophic level.
- Omnivores, parasites, decomposers, detritivores and

scavengers occupy different trophic levels depending on what they happen to be eating at the time.



Trophic Levels and energy flow in Ecosystems

- Each trophic level contains a certain amount of energy.
- As energy flows from one trophic level to the other useful energy is lost, usually as heat to the surroundings- second law of thermodynamics.
- Only about 10% of the chemical energy at one level is transferred to another. 90% of the chemical energy is lost as heat- **Pyramid of energy**
- The percentage of useful energy from one trophic level to another is called **ecological efficiency** or **food chain efficiency**

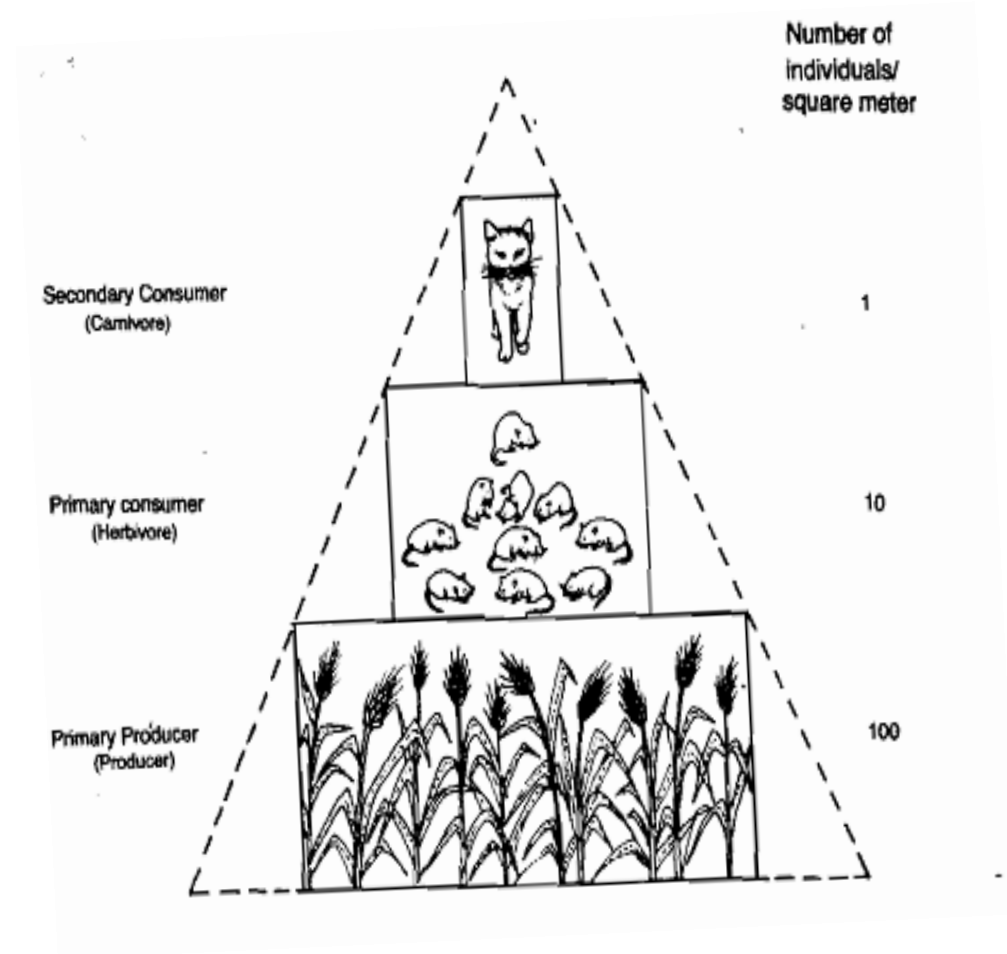


**The shorter the food chain,
the lesser the loss of useful
energy.**

Common Basis for Comparing Ecosystems

❖ Pyramid of Numbers

- The number of organisms in each trophic level is counted
- This basis for comparison is not reliable in most cases.
- If more ants are feeding on one dead animal. The pyramid will be inverted.
- The pyramid of biomass seems to solve the ambiguity in the pyramid of numbers



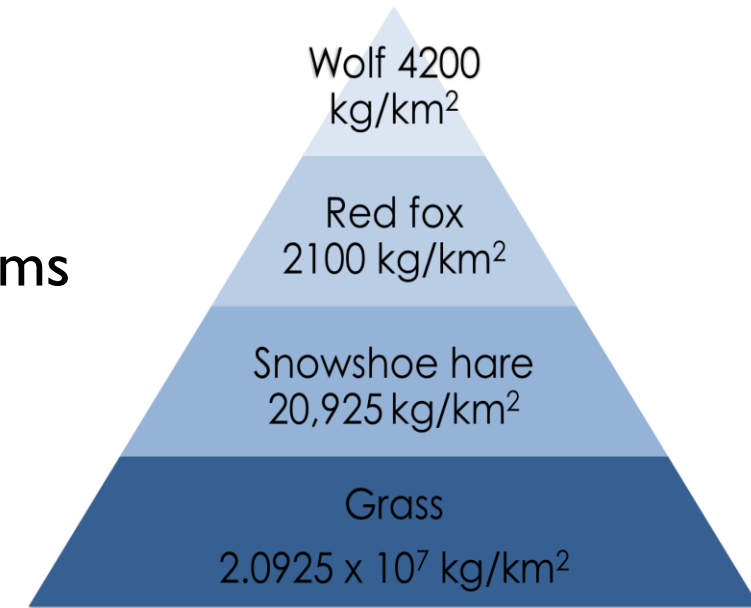
Common Basis for Comparing Ecosystems

❖ Pyramid of the Biomass

- Shows the biomass weight at each trophic level.
- Biomass measurements are based on dry weights of the organisms
- Comparing food webs in fresh weights could be deceptive
- It is usually expressed in g/m² or in metric tons per hectare

❖ Pyramid of Energy

- Energy flow from one trophic level to another is not 100%
- About 10% of the chemical energy available at one trophic level gets transferred to organisms at the next trophic level



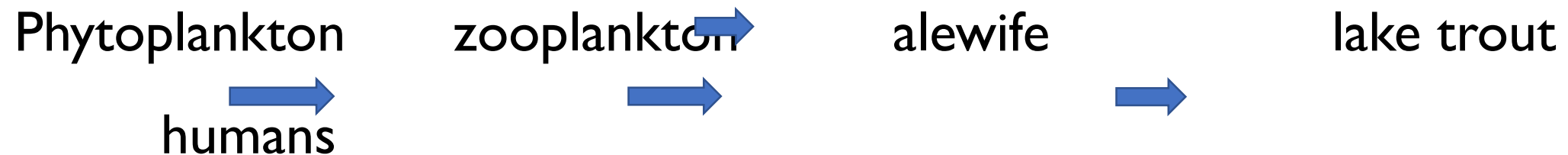
Question

A deer eats 25kg of herbaceous material per day. The herbaceous material is approximately 20% dry matter(DM) and has an energy content of 10MJ. (kg DM)⁻¹. Of the total energy ingested per day, 25% is excreted as undigested material. Of the 75% that is digested, 80% is lost to metabolic waste products and heat. The remaining 20% is converted to body tissue. How MJ are converted to body tissue on a daily basis? Calculate the percentage of energy consumed that is converted to body tissue

- Draw a schematic energy balance

Question

For every megajoule of energy used by the phytoplankton in lake Michigan. How many joules of energy are used in building cell tissue in the lake trout? How many in humans? Use the food chain path



Given only 10% of energy consumed is converted to biomass

Non-living (Abiotic) Portion of an Ecosystem

■ Energy:

Solar energy drives the entire ecosystem by helping to create climate to recycle essential chemicals and to support plant life

■ Physical factors:

- temperature
- light
- wind
- humidity
- current
- rainfall

Non-living (Abiotic) Portion of an Ecosystem

■ Chemical factors:

- Water
- Oxygen
- Carbon dioxide
- Essential minerals and substances (proteins, carbohydrates, lipids, vitamins and other complex chemicals necessary for life)

The critical inorganic and organic chemicals found in air, water and soil must be recycled again several times through the biosphere

Limiting Factors

- Every organism survival depend on chemical and physical factors
- Organisms are affected by a combined action of these factors
- Each organism has a certain range of tolerance to variations in these factors
- A shortage, excess or absence of these factors can affect its number or distribution.
- Such factors are known as limiting factors.
- ❖ It is a resource (nutrient or substance) or environmental condition whose presence, absence, decrease and increase can limit the number growth, distribution, abundance or metabolic processes of an organism.
- ❖ A limiting resource is the nutrient or substance that is in shortest supply in relation to organisms' demand.

The Limiting Factor Principle & Law

- Principle deals with the degree of availability of factors, the distribution and the abundance of organisms.
- The law states that the existence, abundance, or distribution of an organism can be determined by whether the level of one or more limiting factors fall above or below the levels required by the organism.
- Eg. Plants which cannot withstand dry conditions die during dry seasons.

The Limiting Factor Principle & Law

- Limiting factors can be:
- **Environmental conditions:** temperature, sunlight rain, humidity, climate
- **Resources:** food, shelter, light, O₂, CO₂, N, P, Na, salinity, nutrients, soil
- **Biotic / Biological factors:** interactions between organisms such as predation, competition.
- **Human limiting factors:** over population, deforestation, habitat destruction

Kinds of Organism Interactions

Predation

- A type of interaction where one organism (the predator), kills and feeds on the other (prey).
- The predators are adapted and highly specialised for hunting its prey

➤ PREY

- To prevent going extinct. They breed more
- Have some adaption that help them survive.

Kinds of Organism Interactions

Competition:

- In this kind of interaction two organisms strive to obtain the same limited resource.
- Two types of competition
- Intraspecific and Interspecific competition.
- Competition among members of the same species is intraspecific
- Competition between members of different species is referred to interspecific competition.
- E.g. plants with many roots absorb almost all the nitrogen from the soil killing nearby plants.
- Hyenas and vultures compete for carcasses

Kinds of Organism Interactions

- Competition among members of the same spp. is a major force in shaping the evolution of a specie.
- Less well –adapted individuals are more likely to die or be denied mating privileges.
- The successful organisms are likely to have larger numbers of offspring.
- May possess genetic characteristics that are favourable for survival of the species in that particular environment.

Kinds of Organism Interactions

Competitive Exclusion Principle

Is the concept that no two species can occupy the same ecological niche in the same place at the same time.

- The more similar two species are, the more intense will be the competition between them
- If one of the two is better adapted to live in the area than the other
 - The less –fit species must evolve into a slightly different niche.
 - Migrate to a different geographic area or
 - Become extinct.

Kinds of Organism Interactions

Symbiosis

- Is a close long lasting physical/biological relationship between two different species
- At least one of them derives some sort of benefits from the contact / interaction.

Three types of symbiotic relationships exist:

- Parasitism
- Commensalism
- Mutualism.

Kinds of Organism Interactions

Parasitism

- Relationship in which one organism known as the parasite lives in or on another organism known as the host from which it derives some form of benefit e.g. Nourishment. The host is harmed.
- Two kinds of parasites occur
- Endoparasites : parasite lives inside the host. Eg. Tapeworms inside the intestines of animals
- Ectoparasites : parasite lives outside the host. Eg. lice in the hair of humans

Kinds of Organism Interactions

Commensalism

- Relationship between two organisms in which one organism benefits while the other neither benefits nor is harmed.
- May not involve a long-term physical contact.



Kinds of Organism Interactions

Mutualism

- A relationship that benefits both species involved. eg. Bee and flowers
- Obligatory mutualism- both species cannot live without each other.
- Eg. Fungi and lichen. Fungi provides minerals and water for lichen. Lichen uses them to make food for itself and the fungi via photosynthesis
- In other situations, the species can exist separately but are more successful when they are involved in a mutualistic relationship.
- The roots nodules of legumes provide home for certain kinds of soil-nitrogen fixing bacteria for plant growth.
- Similarly, many kinds of fungi form an association with the roots of plants.
- The root-fungus associations are called mycorrhizae.

Climate

- The average atmospheric conditions over a long period of time

Factors that make up climates :

- Light
- temperature,
- precipitation
- humidity

- Five classes of climate

i. Tropical climate

ii. Continental climate

iii. Dry climate

iv. Temperate climate

v. Cold polar climate

The Soil

- A crucial interface between the lithosphere and all living plants that grow on its surface
- Soil is a complex mixture of inorganic and organic materials, water, air and living organisms.

The soil is made of three layers known as horizons

I. The A-Horizon:

- Consists of freshly fallen plant litter and a sub layer of humus made up of decayed matter and some inorganic material, and another sub-layer of insoluble minerals
- Supports life

The Soil

2. The B-Horizon- the subsoil

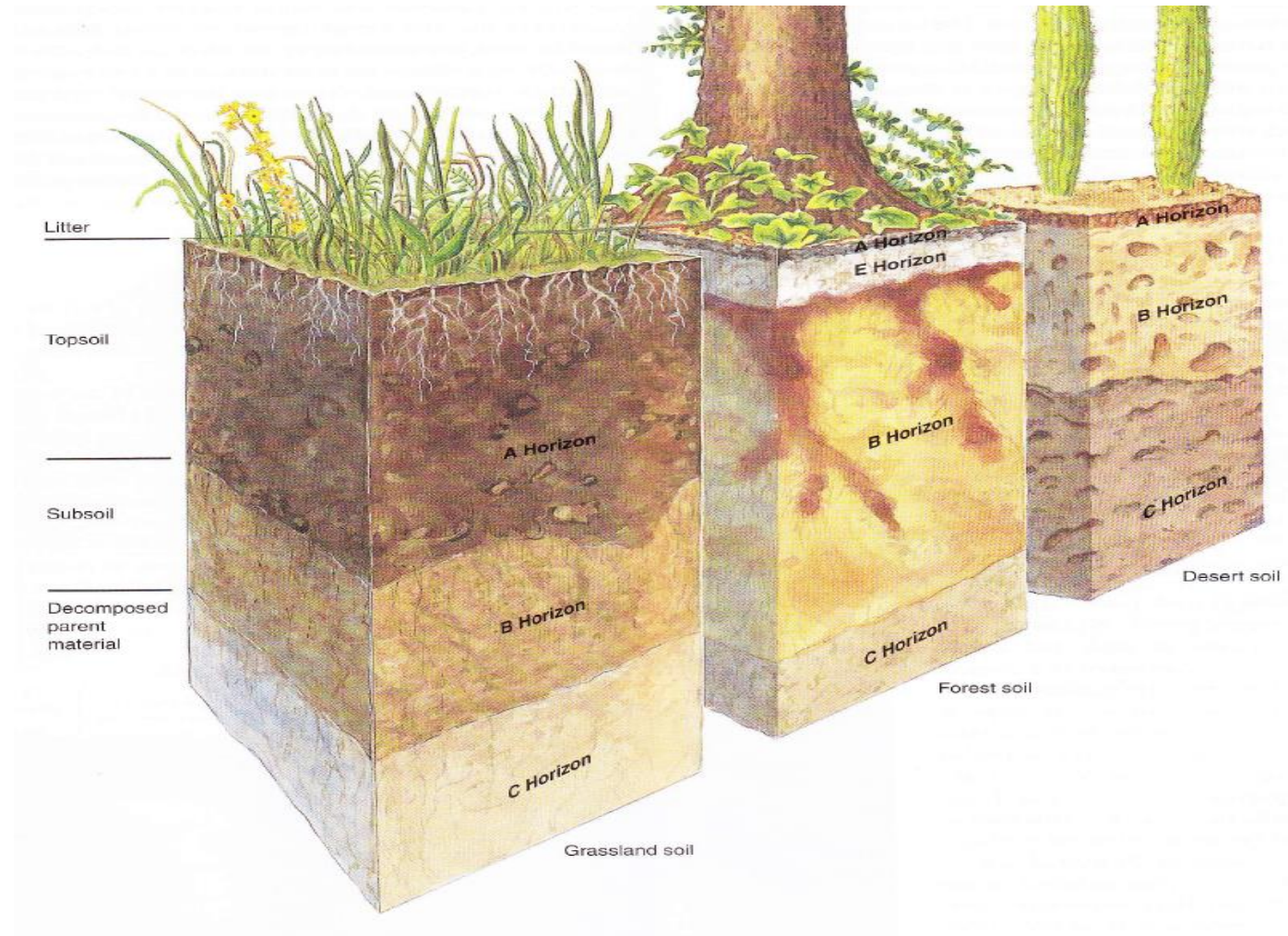
- Contains fine particles of inorganic materials such as clay and some finely divided minerals from the parent rock below
- Contains little organic material. Cannot support life

3. The C-Horizon:

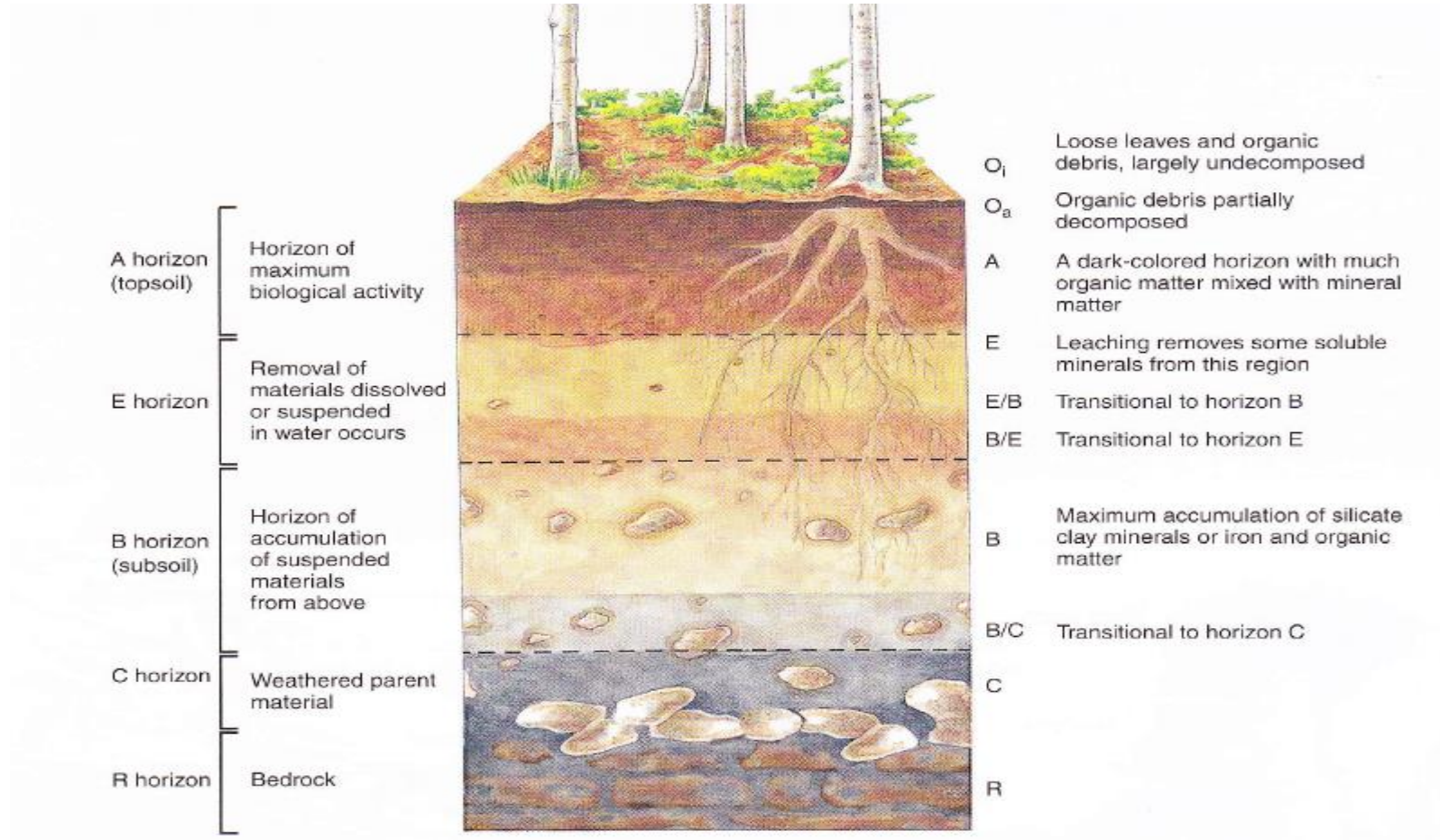
- Contains the pieces of rocks broken down from the parent rock through weather processes.

Different types of soils by ecosystem type

The 3 horizons. of the soils in different biomes in the world vary in colour, depth, physical and chemical characteristics



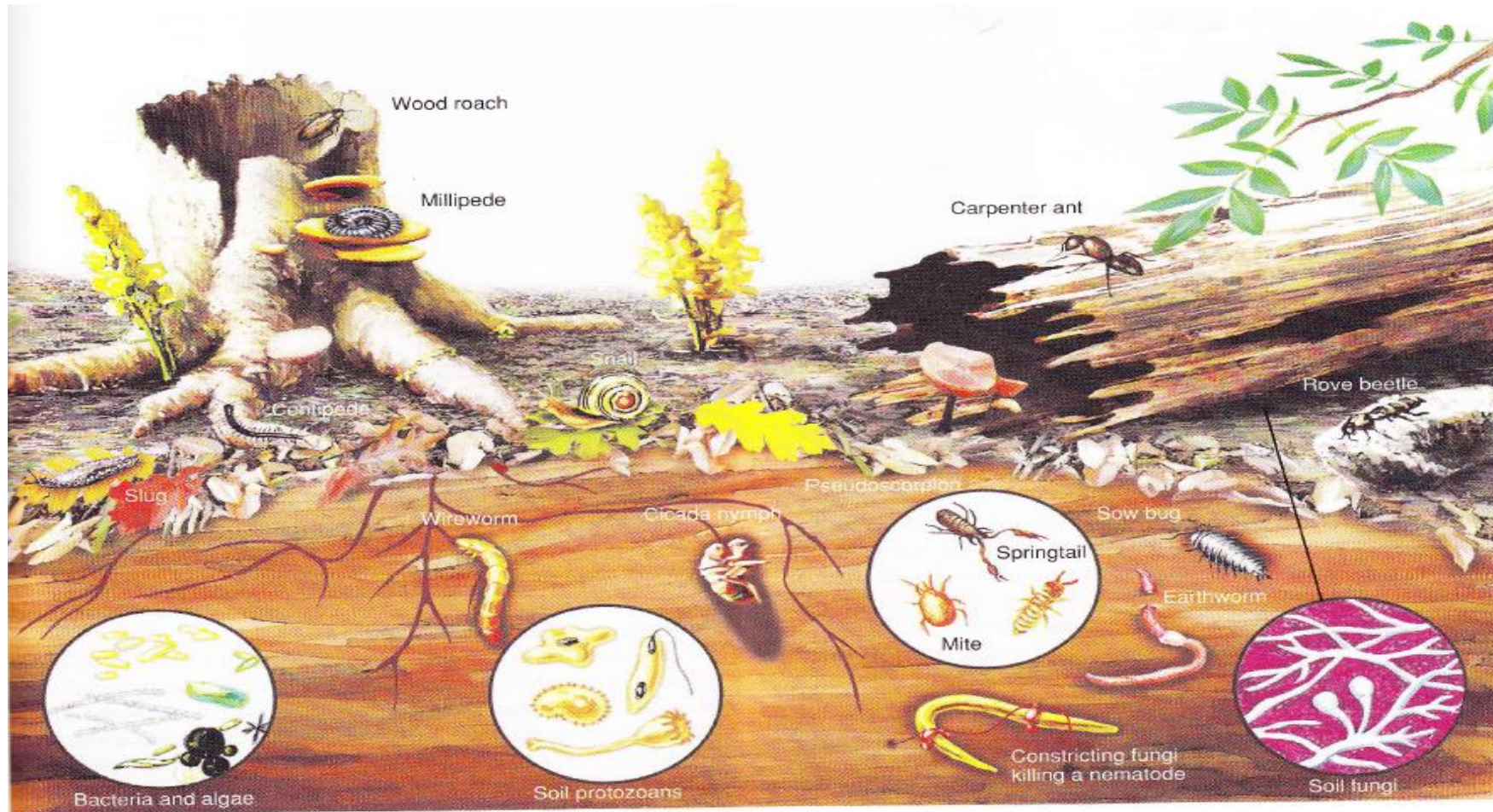
Soil Profile



Relationship between climate, soil and ecosystems

- Climate determine the plant life
- Climate and plants together determine the soil type
- Climate, plants and soil together determine the types, numbers and distribution of animals in a given ecosystem.

Organisms in the soil



Application of pesticides destroy the organisms and weedicides destroy the ecology of the soil

Ecosystem Functions

- The Ecosystem operates through cycling of matter and energy flow.
- Life on earth depends on these 2 ecosystem functions.
- Six elements that are essential for life and make up over 95% of the mass of all living things are: Hydrogen, oxygen, carbon, nitrogen, phosphorus and sulphur.
- Since they are in fixed supply they must be cycled continuously from their reservoir of air, water and soil through the food web and back again into their reservoirs.
- The cycling movements of these chemicals is known as Biogeochemical Cycles

Biogeochemical cycles

- Is a pathway by which conserved matter move through biotic and abiotic components of the ecosystem through the law of conservation of matter
- Abiotic components are: Hydrosphere (water), Atmosphere (air), Lithosphere (rocks)
- Matter cycling in the ecosystem is carried out by natural cycles.

They are 3 types of biogeochemical cycles

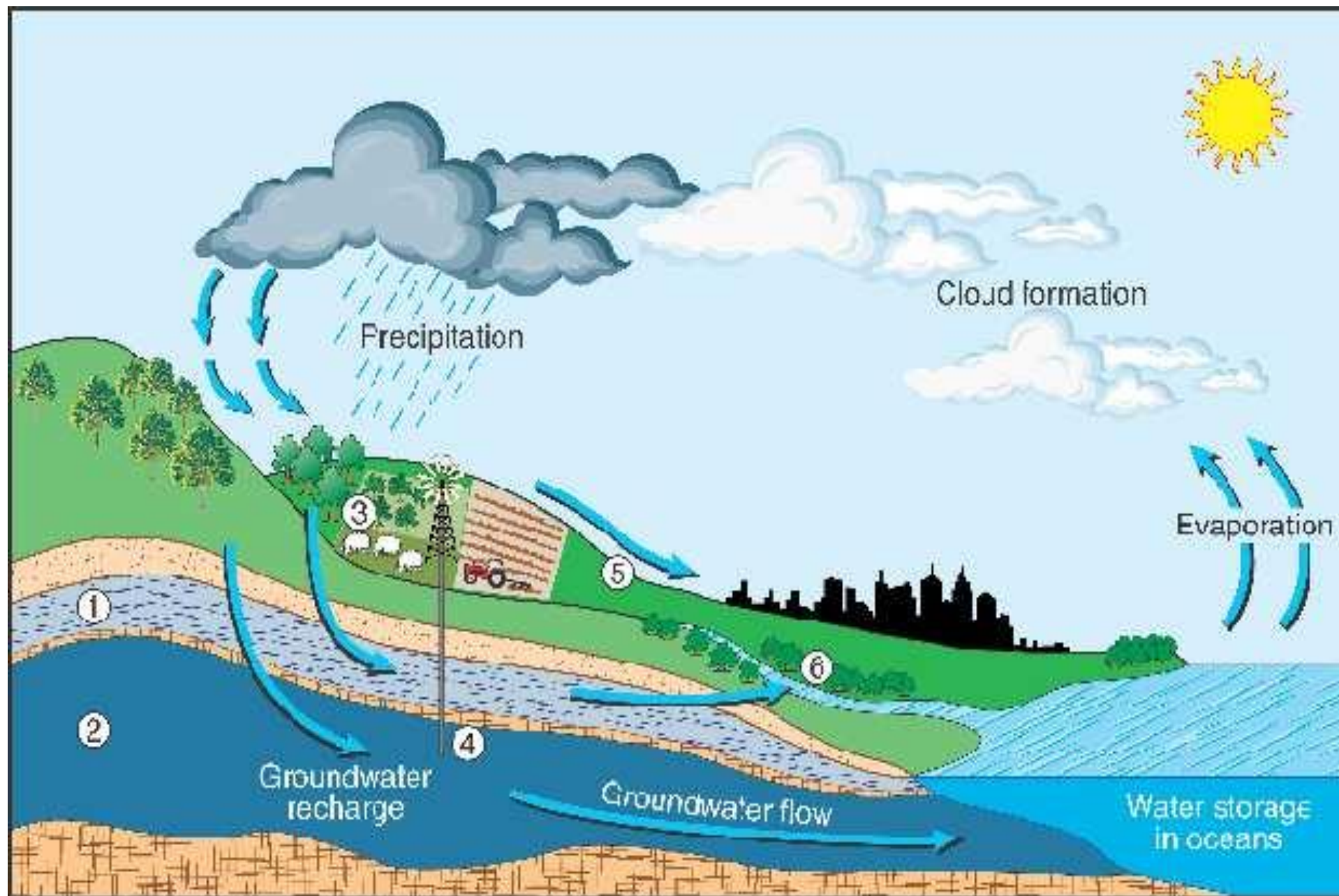
- **Gaseous-** the atmosphere is the major reservoir. Eg. N, C, O₂
- **Hydrological.** Eg. water
- **Sedimentary** cycle – movement of chemicals from land to sea and then back again. Eg. Ca, P, S, Fe, Mg

Hydrological Cycle

- This cycle helps in exchange of water between air, land, sea, living plants and animals.
- One third of the solar radiation absorbed by earth drives the hydrological cycle.
- It involves;
 - i. Massive evaporation of water from the ocean.
 - Evaporation from the ocean exceeds precipitation by 10%. This excess 10% move as water vapour over land surface. They balance the hydrological cycle
 - ii. Cloud formation
 - iii. Depending on the temperature, it may be rain, snow or hail. These gives us our supply and reserves of fresh water

Hydrological Cycle

- When there is a good rainfall, some form run-offs and some percolate into the soil into underground aquifers
- The cycle consist of a balance of constant process of evaporation, transpiration, precipitation, surface run-offs and ground water movement
- **Note:** Aquifers exist above impermeable rock strata
- They are underground reservoirs of water. The water can be extracted.



① Unconfined aquifer

② Confined aquifer

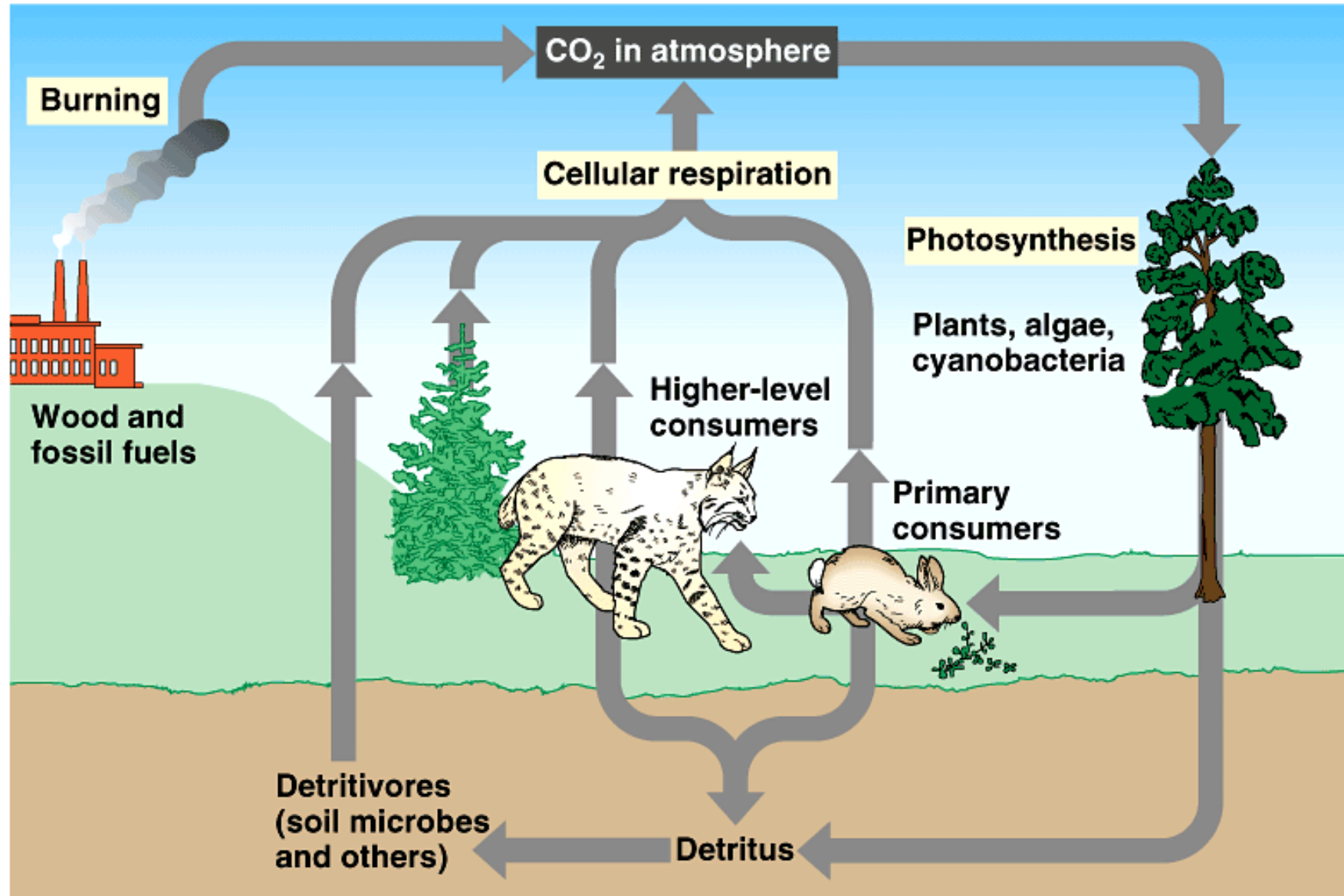
③ Groundwater use

④ Groundwater extraction

⑤ Surface water run off

⑥ Riparian zone

CARBON CYCLE



OXYGEN IN THE ATMOSPHERE

Plants take in oxygen and give out carbon dioxide at night.

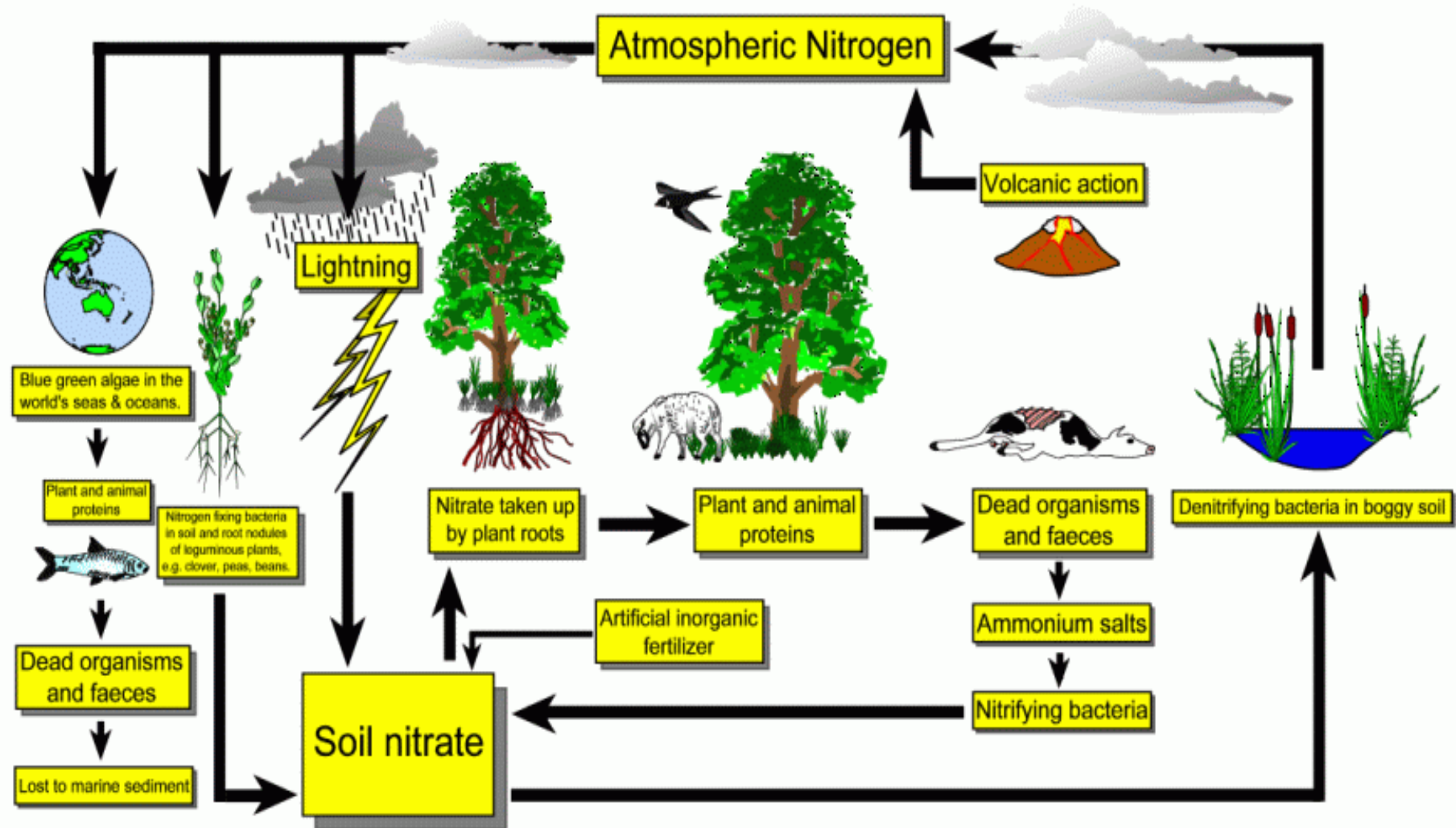
Plants take in carbon dioxide and give oxygen during photosynthesis in the daytime.

Animals breathe in oxygen and breathe out carbon dioxide.

night

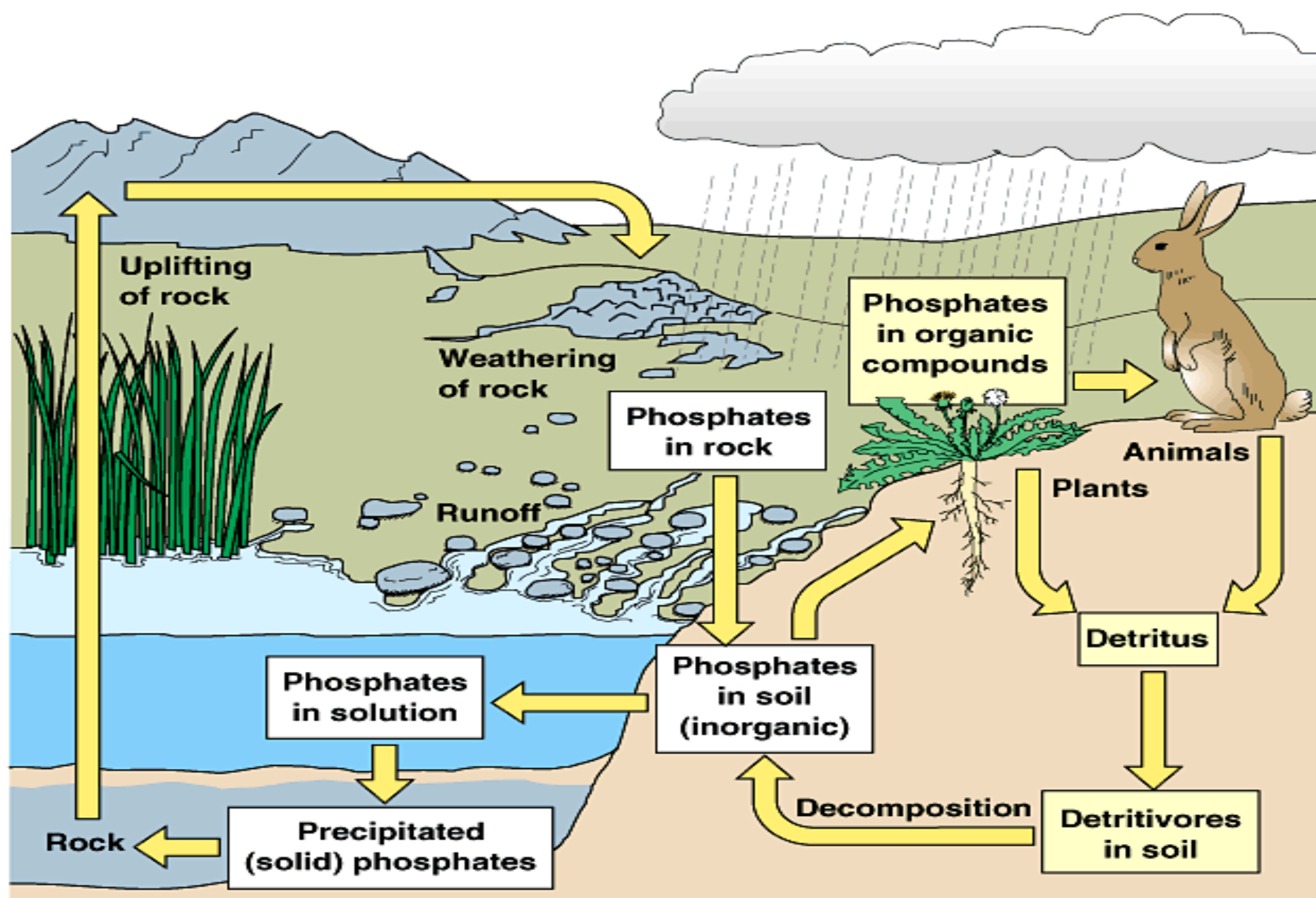
day

The Nitrogen Cycle



Phosphorous Cycle

- Most of elements in the ecosystem follow the concept of the sedimentary cycle where the reservoir is the soil or land. Eg Calcium, Potassium, Phosphorous, Zinc etc
- Attention is focused on phosphorous because it is becoming a limiting factor in the ecosystems
- Phosphorous is not an abundant component of the ecosphere
- The earth is the major reservoir of phosphorous
- Phosphorous is slowly recycled from land to sea and back to the land
- Phosphate is normally reduced by micro-organisms
- The world as a whole will not run out of phosphorous for a long time but local and regional shortages already exists and could be getting worse



Assignment I

- Develop an *Earthmanship City* which is based on matter cycling, effective social organization, pollution control, energy conservation co-operation with each other and the environment.
- Describe how urban systems and natural ecosystems differ and their similarities. Debate the idea that private landowners have the right to do anything they want on their land. How is land-use decided in your community? What role do citizens have? What type of ethics is practiced? On the whole has this been on behalf of economic interests or ecological interests? From the discussion what is the future of the environment in Ghana?

Assignment 2

- 1. Describe the functions of the soil to the ecosystem
- 2. What is a niche in an ecosystem?
- 3. What is your niche in the home?
- 4. If the major reservoir of this element is the a) soil b) atmosphere c) water then what type of cycle is that?
- 5. What is a limiting factor in an ecosystem
- 6. Is the city of Accra an ecosystem? What will happen if there is a total lockdown with no flow of matter from external sources? How long will the human being last without food?